Signal/background separation · mk3 · round 1

```
date         2026-10-25 17:42
data         C:\Users\natha\OneDrive\Desktop\0_mL_ATLAS\0_Sig_BG_09_21_KJ\kj_092126\run3_csv_for_nathan
background   DataHTMissLow_train_data24_cleaned.csv  rows 36,059
signal       HSS_mH125_mS15_ct580_cleaned.csv  rows 570
signal       HSS_mH125_mS55_ct5320_cleaned.csv  rows 423
signal       HSS_mH200_mS50_ct1255_cleaned.csv  rows 1,520
signal       HSS_mH400_mS100_ct1610_cleaned.csv  rows 3,792
signal       HSS_mH600_mS150_ct3310_cleaned.csv  rows 4,490
signal       HSS_mH1000_mS275_ct2399_cleaned.csv  rows 4,188
region       ('msvtx_inBarrel_NOSYS', 'flag', 'msvtx_inEndcaps_NOSYS')
NN lists     C:\Users\natha\OneDrive\Desktop\0_Fall_26\0_Oct_26_runs\kj_DashBoard_mk\nn_features_mk4.txt

unit test    HTmiss selection: PASS  (details next page)

features common to all 7 files     230
hard / user excluded               15 / 0
non-numeric / constant overall     6 / 20
rank-identical duplicates dropped  91  (in 41 groups; duplicates.txt)
features scored                    98

KS D = sup|F_s - F_b| in [0, 1] = best single-cut (signal eff - background eff).
noise floor D_crit = 1.36 sqrt(1/n_s + 1/n_b), 95% level for identical distributions.
D_equal weights each mass point 1/N_m (n_s = effective size); D_pooled concatenates them.
folders: features/NNN_<feature>/, NNN = rank by endcap D_equal

pages: cover, unit test, NN feature table, strong-but-unused features, then per region a
ranking chart (tick = noise floor) and a top-40 table.
```

Unit test

```
================================================================================
UNIT TEST  HTmiss selection:  background htmiss_NOSYS < 40 GeV,  signal htmiss_NOSYS >= 40 GeV
================================================================================
units: MeV (signal median 158.5 GeV)
  file                n invalid   min GeV   max GeV violating  result
  bkg            36,059       0      0.00     40.00         0  PASS
  mH125_mS15        570       0     40.36    768.98         0  PASS
  mH125_mS55        423       0     40.05    648.06         0  PASS
  mH200_mS50      1,520       0     40.08   1101.73         0  PASS
  mH400_mS100     3,792       0     40.20   1191.45         0  PASS
  mH600_mS150     4,490       0     40.00   1321.18         0  PASS
  mH1000_mS275    4,188       0     40.02   1447.43         0  PASS

RESULT: PASS -- background and signal sit on opposite sides of the cut, as designed
```

NN feature lists: KS D in each list's region (1/2)

NN1 barrel (20 features, scored in region barrel)

list name	D_equal	noise	min	max	rank	scored as
MS1Vtx_mindR_jetcut	0.365	0.051	0.307	0.432	23/98	MS1Vtx_mindR_jetcut_raw [region-tagged (barrel): MS1Vtx_barrel_mindR_jetcut]
msegUnAssoc_counts/nBOL	0.319	0.051	0.142	0.508	28/98	nBOL_NOSYS [exact: msegUnAssoc_counts/nBOL]
MS1Vtx_maxclusE	0.312	0.051	0.275	0.332	31/98	MS1Vtx_maxclusE_raw [region-tagged (barrel): MS1Vtx_barrel_maxclusE]
MS1Vtx_clusE	0.310	0.051	0.282	0.343	32/98	msvtx_caloClusterSumE [region-tagged (barrel): MS1Vtx_barrel_clusE]
msvtx_caloClusterSumPtScalar	0.307	0.051	0.284	0.347	34/98	msvtx_caloClusterSumPtScalar
MS1Vtx_avg_clusE	0.304	0.051	0.252	0.333	35/98	MS1Vtx_avg_clusE_raw [region-tagged (barrel): MS1Vtx_barrel_avg_clusE]
MS1Vtx_rms_clusE	0.299	0.051	0.261	0.331	36/98	MS1Vtx_rms_clusE_raw [region-tagged (barrel): MS1Vtx_barrel_rms_clusE]
MS1Vtx_l3ecal	0.293	0.051	0.251	0.331	38/98	msvtx_el3ecal [region-tagged (barrel): MS1Vtx_barrel_l3ecal]
msegUnAssoc_counts/nBML	0.277	0.051	0.053	0.473	44/98	msegUnAssoc_counts/nBML
MS1Vtx_l2ecal	0.267	0.051	0.218	0.321	47/98	msvtx_el2ecal [region-tagged (barrel): MS1Vtx_barrel_l2ecal]
track_scalar_sum_pt_barrel	0.265	0.051	0.183	0.327	48/98	track_scalar_sum_pt_barrel
nMSeg_ratio_BIBM	0.263	0.051	0.242	0.313	49/98	nMSeg_ratio_BIBM
MS1Vtx_l2hcal	0.259	0.051	0.239	0.293	51/98	msvtx_el2hcal [region-tagged (barrel): MS1Vtx_barrel_l2hcal]
MS1Vtx_l4ecal	0.243	0.051	0.201	0.272	52/98	msvtx_el4ecal [region-tagged (barrel): MS1Vtx_barrel_l4ecal]
MS1Vtx_l3hcal	0.240	0.051	0.224	0.265	53/98	msvtx_el3hcal [region-tagged (barrel): MS1Vtx_barrel_l3hcal]
MS1Vtx_l1ecal	0.235	0.051	0.187	0.267	55/98	msvtx_el1ecal [region-tagged (barrel): MS1Vtx_barrel_l1ecal]
MS1Vtx_l4hcal	0.166	0.051	0.139	0.202	61/98	msvtx_el4hcal [region-tagged (barrel): MS1Vtx_barrel_l4hcal]
msegUnAssoc_counts/nBIL	0.152	0.051	0.102	0.243	65/98	msegUnAssoc_counts/nBIL
MS1Vtx_sumTrackPt0p2Cone	0.111	0.051	0.092	0.119	73/98	msvtx_trackSoftSumPt_NOSYS [region-tagged (barrel): MS1Vtx_barrel_sumTrackPt0p2Cone]
MS1Vtx_l1hcal	nan	0.051	nan	nan	98/98	msvtx_el1hcal [region-tagged (barrel): MS1Vtx_barrel_l1hcal]

NN1 endcap (16 features, scored in region endcap)

list name	D_equal	noise	min	max	rank	scored as
MS1Vtx_clusE	0.383	0.022	0.372	0.396	26/98	msvtx_caloClusterSumE [region-tagged (endcap): MS1Vtx_endcap_clusE]
MS1Vtx_maxclusE	0.374	0.022	0.351	0.386	29/98	MS1Vtx_maxclusE_raw [region-tagged (endcap): MS1Vtx_endcap_maxclusE]
MS1Vtx_rms_clusE	0.365	0.022	0.342	0.384	32/98	MS1Vtx_rms_clusE_raw [region-tagged (endcap): MS1Vtx_endcap_rms_clusE]
MS1Vtx_l3ecal	0.364	0.022	0.349	0.382	34/98	msvtx_el3ecal [region-tagged (endcap): MS1Vtx_endcap_l3ecal]
MS1Vtx_avg_clusE	0.361	0.022	0.336	0.382	35/98	MS1Vtx_avg_clusE_raw [region-tagged (endcap): MS1Vtx_endcap_avg_clusE]
MS1Vtx_l2ecal	0.343	0.022	0.336	0.363	38/98	msvtx_el2ecal [region-tagged (endcap): MS1Vtx_endcap_l2ecal]
MS1Vtx_l4ecal	0.337	0.022	0.293	0.368	39/98	msvtx_el4ecal [region-tagged (endcap): MS1Vtx_endcap_l4ecal]
MS1Vtx_l1ecal	0.324	0.022	0.307	0.337	42/98	msvtx_el1ecal [region-tagged (endcap): MS1Vtx_endcap_l1ecal]
MS1Vtx_l1hcal	0.313	0.022	0.288	0.331	43/98	msvtx_el1hcal [region-tagged (endcap): MS1Vtx_endcap_l1hcal]
MS1Vtx_l2hcal	0.311	0.022	0.299	0.323	45/98	msvtx_el2hcal [region-tagged (endcap): MS1Vtx_endcap_l2hcal]
MS1Vtx_mindR_jetcut	0.308	0.022	0.259	0.356	46/98	MS1Vtx_mindR_jetcut_raw [region-tagged (endcap): MS1Vtx_endcap_mindR_jetcut]
track_scalar_sum_pt_endcap	0.286	0.022	0.251	0.347	48/98	track_scalar_sum_pt_endcap
MS1Vtx_l3hcal	0.193	0.022	0.184	0.197	59/98	msvtx_el3hcal [region-tagged (endcap): MS1Vtx_endcap_l3hcal]
MS1Vtx_sumTrackPt0p2Cone	0.140	0.022	0.136	0.143	65/98	msvtx_trackSoftSumPt_NOSYS [region-tagged (endcap): MS1Vtx_endcap_sumTrackPt0p2Cone]
MS1Vtx_l4hcal	0.094	0.022	0.080	0.110	79/98	msvtx_el4hcal [region-tagged (endcap): MS1Vtx_endcap_l4hcal]
nMSeg_ratio_EIEM	0.086	0.022	0.078	0.177	81/98	nMSeg_ratio_EIEM

NN2 barrel (10 features, scored in region barrel)

list name	D_equal	noise	min	max	rank	scored as
MS1Vtx_barrel_hits_ntot	0.632	0.051	0.405	0.834	13/98	MS1Vtx_barrel_hits_ntot
MS1Vtx_nTracklet	0.442	0.051	0.418	0.475	17/98	msvtx_nTrk [region-tagged (barrel): MS1Vtx_barrel_nTracklet]
MSVtx_MET_dphi	0.429	0.051	0.332	0.620	19/98	MSVtx_MET_dphi_raw [region-tagged (barrel): MSVtx_MET_barrel_dphi]
MSeg_barrel_avg_dR	0.366	0.051	0.101	0.645	22/98	MSeg_barrel_avg_dR
MSeg_barrel_rms_dR	0.297	0.051	0.133	0.584	37/98	MSeg_barrel_rms_dR
MS1Vtx_maxclustime	0.274	0.051	0.221	0.322	45/98	MS1Vtx_maxclustime_raw [region-tagged (barrel): MS1Vtx_barrel_maxclustime]
MS1Vtx_rms_clustime	0.262	0.051	0.218	0.317	50/98	MS1Vtx_rms_clustime_raw [region-tagged (barrel): MS1Vtx_barrel_rms_clustime]
MS1Vtx_avg_clustime	0.157	0.051	0.108	0.199	63/98	MS1Vtx_avg_clustime_raw [region-tagged (barrel): MS1Vtx_barrel_avg_clustime]
MSTracklet_barrel_rms_dR	0.097	0.051	0.066	0.150	78/98	MSTracklet_barrel_rms_dR
MSTracklet_barrel_avg_dR	0.097	0.051	0.067	0.150	77/98	MSTracklet_barrel_avg_dR

NN2 endcap (10 features, scored in region endcap)

list name	D_equal	noise	min	max	rank	scored as

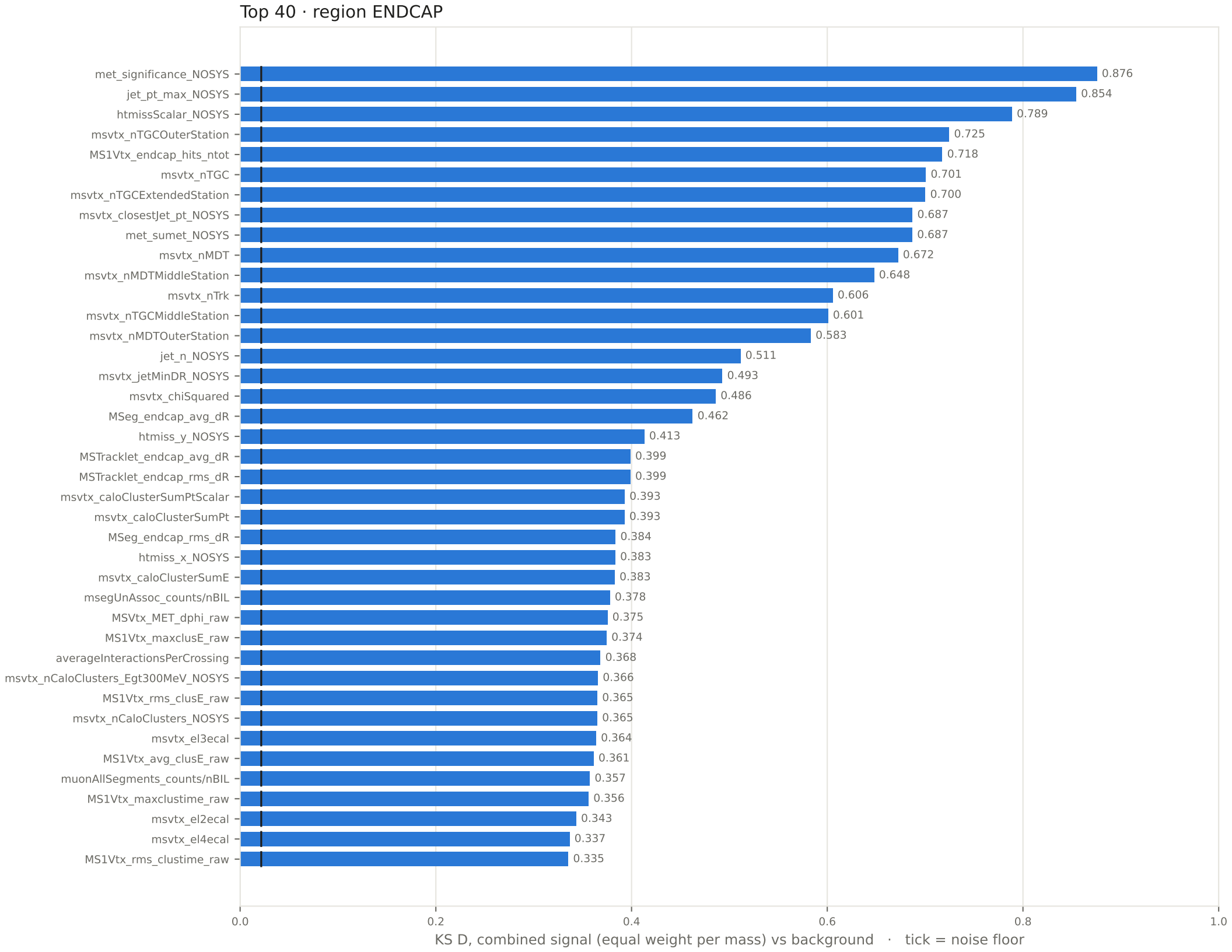
NN feature lists: KS D in each list's region (2/2)

```
--------------------------------------------------------------------------------
MS1Vtx_endcap_hits_ntot     0.718   0.022   0.553   0.864      5/98  MS1Vtx_endcap_hits_ntot
MS1Vtx_nTracklet            0.606   0.022   0.551   0.643     12/98  msvtx_nTrk  [region-tagged (endcap): MS1Vtx_endcap_nTracklet]
MSeg_endcap_avg_dR          0.462   0.022   0.235   0.667     18/98  MSeg_endcap_avg_dR
MSTracklet_endcap_avg_dR    0.399   0.022   0.297   0.502     20/98  MSTracklet_endcap_avg_dR
MSTracklet_endcap_rms_dR    0.399   0.022   0.297   0.502     21/98  MSTracklet_endcap_rms_dR
MSeg_endcap_rms_dR          0.384   0.022   0.180   0.594     24/98  MSeg_endcap_rms_dR
MSVtx_MET_dphi              0.375   0.022   0.273   0.508     28/98  MSVtx_MET_dphi_raw  [region-tagged (endcap): MSVtx_MET_endcap_dphi]
MS1Vtx_maxclustime          0.356   0.022   0.345   0.371     37/98  MS1Vtx_maxclustime_raw  [region-tagged (endcap): MS1Vtx_endcap_maxclustime]
MS1Vtx_rms_clustime         0.335   0.022   0.309   0.356     40/98  MS1Vtx_rms_clustime_raw  [region-tagged (endcap): MS1Vtx_endcap_rms_clustime]
MS1Vtx_avg_clustime         0.229   0.022   0.218   0.238     55/98  MS1Vtx_avg_clustime_raw  [region-tagged (endcap): MS1Vtx_endcap_avg_clustime]
```

Strong features not in the NN lists

```
ENDCAP: top 15 features used by neither NN1 nor NN2 endcap
    1  met_significance_NOSYS                   D_equal 0.876  (min 0.810, max 0.939)
    2  jet_pt_max_NOSYS                         D_equal 0.854  (min 0.785, max 0.920)
    3  htmissScalar_NOSYS                       D_equal 0.789  (min 0.789, max 0.868)
    4  msvtx_nTGCOuterStation                   D_equal 0.725  (min 0.612, max 0.839)
    6  msvtx_nTGC                               D_equal 0.701  (min 0.590, max 0.840)
    7  msvtx_nTGCExtendedStation                D_equal 0.700  (min 0.585, max 0.828)
    8  msvtx_closestJet_pt_NOSYS                D_equal 0.687  (min 0.647, max 0.715)
    9  met_sumet_NOSYS                          D_equal 0.687  (min 0.563, max 0.859)
   10  msvtx_nMDT                               D_equal 0.672  (min 0.491, max 0.846)
   11  msvtx_nMDTMiddleStation                  D_equal 0.648  (min 0.504, max 0.827)
   13  msvtx_nTGCMiddleStation                  D_equal 0.601  (min 0.471, max 0.755)
   14  msvtx_nMDTOuterStation                   D_equal 0.583  (min 0.465, max 0.686)
   15  jet_n_NOSYS                              D_equal 0.511  (min 0.423, max 0.647)
   16  msvtx_jetMinDR_NOSYS                     D_equal 0.493  (min 0.468, max 0.538)
   17  msvtx_chiSquared                         D_equal 0.486  (min 0.582, max 1.000)

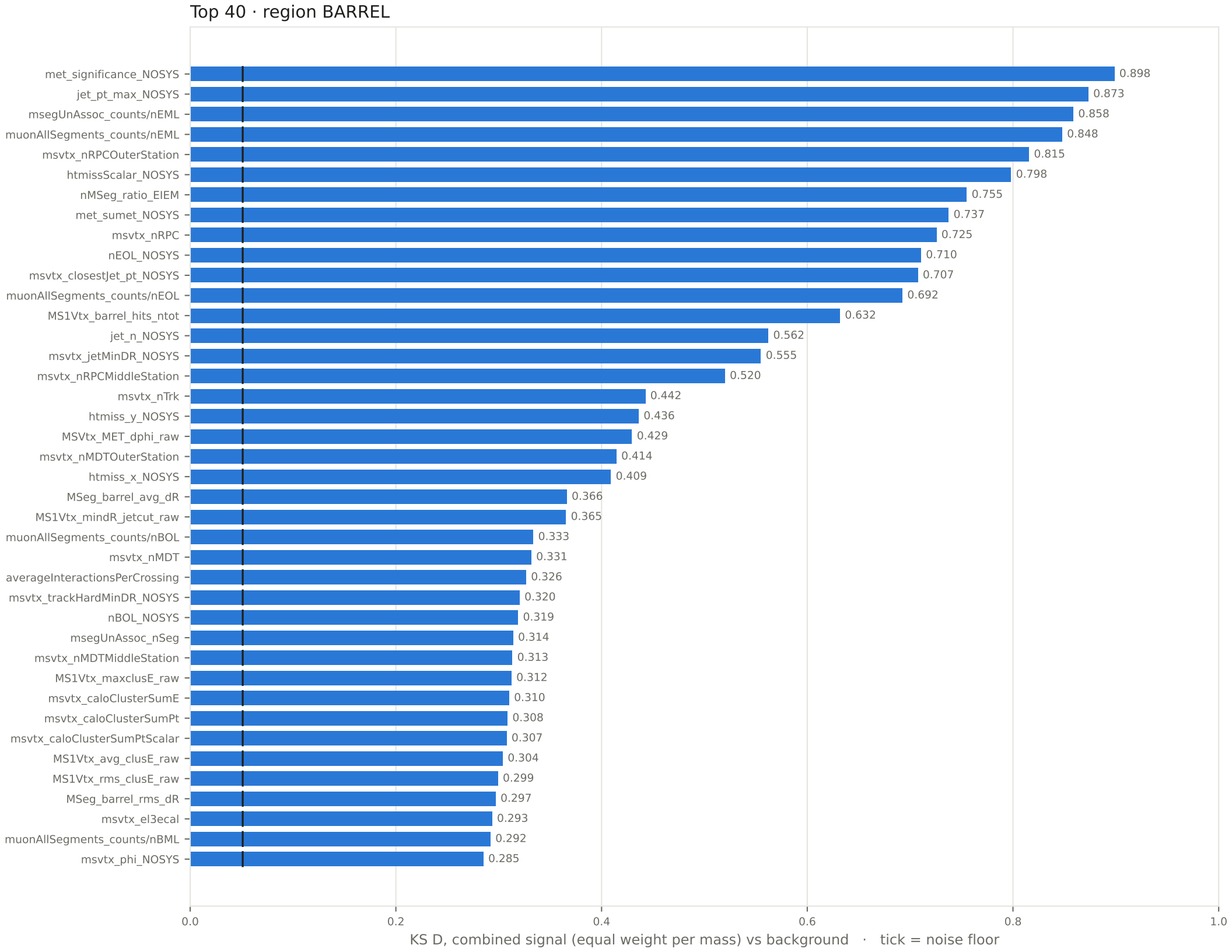
BARREL: top 15 features used by neither NN1 nor NN2 barrel
    1  met_significance_NOSYS                   D_equal 0.898  (min 0.831, max 0.962)
    2  jet_pt_max_NOSYS                         D_equal 0.873  (min 0.789, max 0.916)
    3  msegUnAssoc_counts/nEML                  D_equal 0.858  (min 0.821, max 0.894)
    4  muonAllSegments_counts/nEML              D_equal 0.848  (min 0.799, max 0.901)
    5  msvtx_nRPCOuterStation                   D_equal 0.815  (min 0.740, max 0.881)
    6  htmissScalar_NOSYS                       D_equal 0.798  (min 0.798, max 0.872)
    7  nMSeg_ratio_EIEM                         D_equal 0.755  (min 0.727, max 0.805)
    8  met_sumet_NOSYS                          D_equal 0.737  (min 0.588, max 0.852)
    9  msvtx_nRPC                               D_equal 0.725  (min 0.647, max 0.850)
   10  nEOL_NOSYS                               D_equal 0.710  (min 0.580, max 0.844)
   11  msvtx_closestJet_pt_NOSYS                D_equal 0.707  (min 0.639, max 0.740)
   12  muonAllSegments_counts/nEOL              D_equal 0.692  (min 0.563, max 0.821)
   14  jet_n_NOSYS                              D_equal 0.562  (min 0.458, max 0.691)
   15  msvtx_jetMinDR_NOSYS                     D_equal 0.555  (min 0.510, max 0.622)
   16  msvtx_nRPCMiddleStation                  D_equal 0.520  (min 0.314, max 0.691)
```


Top 40 · region ENDCAP
met_significance_NOSYS 0.876
jet_pt_max_NOSYS 0.854
htmissScalar_NOSYS 0.789
msvtx_nTGCOuterStation 0.725
MS1Vtx_endcap_hits_ntot 0.718
msvtx_nTGC 0.701
msvtx_nTGCExtendedStation 0.700
msvtx_closestJet_pt_NOSYS 0.687
met_sumet_NOSYS 0.687
msvtx_nMDT 0.672
msvtx_nMDTMiddleStation 0.648
msvtx_nTrk 0.606
msvtx_nTGCMiddleStation 0.601
msvtx_nMDTOuterStation 0.583
jet_n_NOSYS 0.511
msvtx_jetMinDR_NOSYS 0.493
msvtx_chiSquared 0.486
MSeg_endcap_avg_dR 0.462
htmiss_y_NOSYS 0.413
MSTracklet_endcap_avg_dR 0.399
MSTracklet_endcap_rms_dR 0.399
msvtx_caloClusterSumPtScalar 0.393
msvtx_caloClusterSumPt 0.393
MSeg_endcap_rms_dR 0.384
htmiss_x_NOSYS 0.383
msvtx_caloClusterSumE 0.383
msegUnAssoc_counts/nBIL 0.378
MSVtx_MET_dphi_raw 0.375
MS1Vtx_maxclusE_raw 0.374
averageInteractionsPerCrossing 0.368
msvtx_nCaloClusters_Egt300MeV_NOSYS 0.366
MS1Vtx_rms_clusE_raw 0.365
msvtx_nCaloClusters_NOSYS 0.365
msvtx_el3ecal 0.364
MS1Vtx_avg_clusE_raw 0.361
muonAllSegments_counts/nBIL 0.357
MS1Vtx_maxclustime_raw 0.356
msvtx_el2ecal 0.343
msvtx_el4ecal 0.337
MS1Vtx_rms_clustime_raw 0.335
0.0
0.2
0.4
0.6
0.8
1.0
KS D, combined signal (equal weight per mass) vs background · tick = noise floor

Top 40 table · region ENDCAP

#	feature	D_equal	noise	D_pooled	D_min	D_max	n_bkg
1	met_significance_NOSYS	0.876	0.022	0.896	0.810	0.939	34,970
2	jet_pt_max_NOSYS	0.854	0.022	0.874	0.785	0.920	34,970
3	htmissScalar_NOSYS	0.789	0.022	0.789	0.789	0.868	34,970
4	msvtx_nTGCOuterStation	0.725	0.022	0.785	0.612	0.839	34,970
5	MS1Vtx_endcap_hits_ntot	0.718	0.022	0.794	0.553	0.864	34,970
6	msvtx_nTGC	0.701	0.022	0.773	0.590	0.840	34,970
7	msvtx_nTGCExtendedStation	0.700	0.022	0.766	0.585	0.828	34,970
8	msvtx_closestJet_pt_NOSYS	0.687	0.022	0.697	0.647	0.715	34,970
9	met_sumet_NOSYS	0.687	0.022	0.741	0.563	0.859	34,970
10	msvtx_nMDT	0.672	0.022	0.759	0.491	0.846	34,970
11	msvtx_nMDTMiddleStation	0.648	0.022	0.736	0.504	0.827	34,970
12	msvtx_nTrk	0.606	0.022	0.607	0.551	0.643	34,970
13	msvtx_nTGCMiddleStation	0.601	0.022	0.683	0.471	0.755	34,970
14	msvtx_nMDTOuterStation	0.583	0.022	0.636	0.465	0.686	34,970
15	jet_n_NOSYS	0.511	0.022	0.556	0.423	0.647	34,970
16	msvtx_jetMinDR_NOSYS	0.493	0.022	0.512	0.468	0.538	34,970
17	msvtx_chiSquared	0.486	0.022	0.576	0.582	1.000	34,970
18	MSeg_endcap_avg_dR	0.462	0.022	0.541	0.235	0.667	34,970
19	htmiss_y_NOSYS	0.413	0.022	0.417	0.368	0.449	34,970
20	MSTracklet_endcap_avg_dR	0.399	0.022	0.431	0.297	0.502	34,970
21	MSTracklet_endcap_rms_dR	0.399	0.022	0.431	0.297	0.502	34,970
22	msvtx_caloClusterSumPtScalar	0.393	0.022	0.394	0.389	0.404	34,970
23	msvtx_caloClusterSumPt	0.393	0.022	0.393	0.388	0.405	34,970
24	MSeg_endcap_rms_dR	0.384	0.022	0.457	0.180	0.594	34,970
25	htmiss_x_NOSYS	0.383	0.022	0.398	0.354	0.438	34,970
26	msvtx_caloClusterSumE	0.383	0.022	0.386	0.372	0.396	34,970
27	msegUnAssoc_counts/nBIL	0.378	0.022	0.346	0.314	0.447	34,970
28	MSVtx_MET_dphi_raw	0.375	0.022	0.357	0.273	0.508	34,970
29	MS1Vtx_maxclusE_raw	0.374	0.022	0.378	0.351	0.386	34,970
30	averageInteractionsPerCrossing	0.368	0.022	0.373	0.338	0.391	34,970
31	msvtx_nCaloClusters_Egt300MeV_NOSYS	0.366	0.022	0.365	0.355	0.375	34,970
32	MS1Vtx_rms_clusE_raw	0.365	0.022	0.372	0.342	0.384	34,970
33	msvtx_nCaloClusters_NOSYS	0.365	0.022	0.364	0.356	0.375	34,970
34	msvtx_el3ecal	0.364	0.022	0.366	0.349	0.382	34,970
35	MS1Vtx_avg_clusE_raw	0.361	0.022	0.369	0.336	0.382	34,970
36	muonAllSegments_counts/nBIL	0.357	0.022	0.319	0.269	0.426	34,970
37	MS1Vtx_maxclustime_raw	0.356	0.022	0.358	0.345	0.371	34,970
38	msvtx_el2ecal	0.343	0.022	0.349	0.336	0.363	34,970
39	msvtx_el4ecal	0.337	0.022	0.340	0.293	0.368	34,970
40	MS1Vtx_rms_clustime_raw	0.335	0.022	0.341	0.309	0.356	34,970

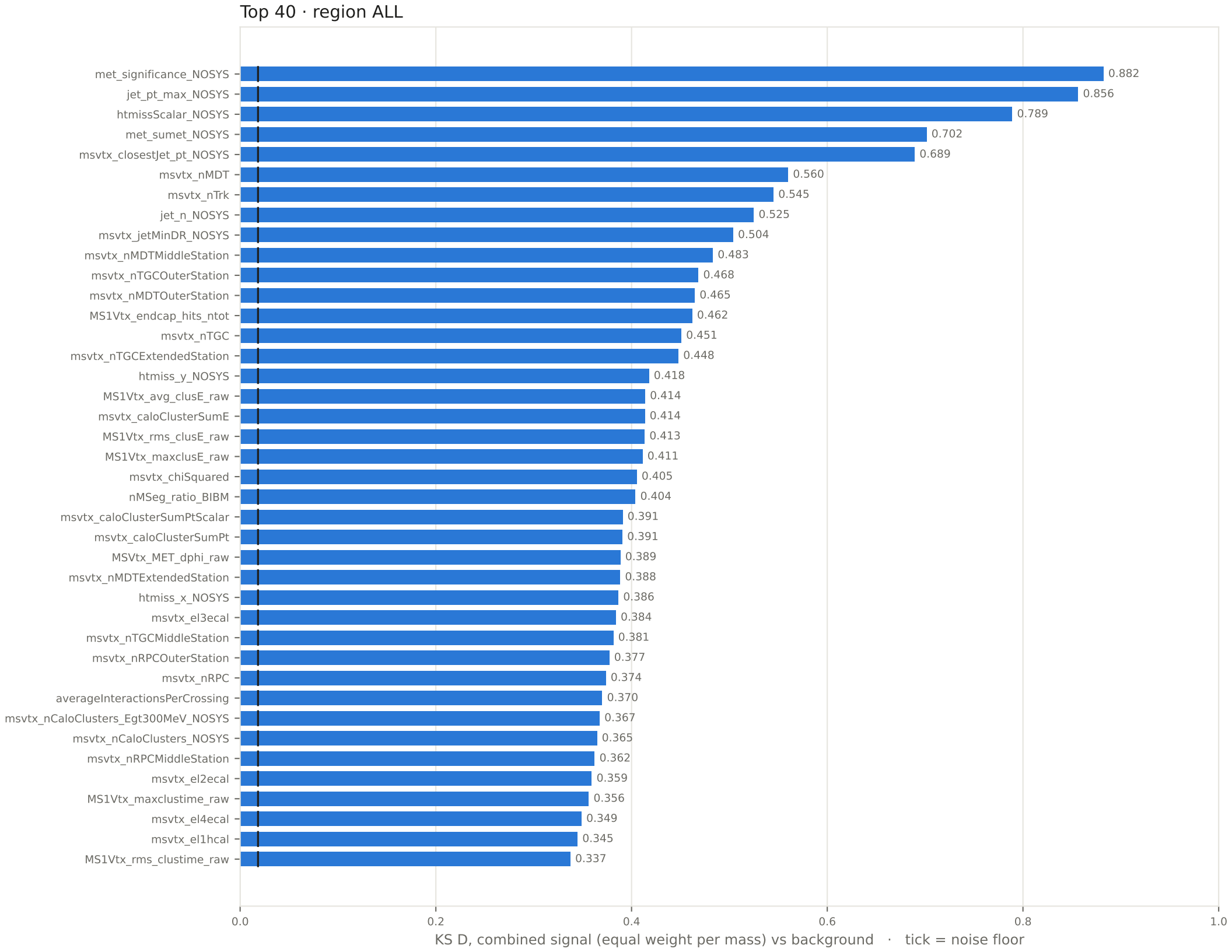
* = D_equal below its noise floor (not distinguishable from no separation)


Top 40 · region BARREL
met_significance_NOSYS 0.898
jet_pt_max_NOSYS 0.873
msegUnAssoc_counts/nEML 0.858
muonAllSegments_counts/nEML 0.848
msvtx_nRPCOuterStation 0.815
htmissScalar_NOSYS 0.798
nMSeg_ratio_EIEM 0.755
met_sumet_NOSYS 0.737
msvtx_nRPC 0.725
nEOL_NOSYS 0.710
msvtx_closestJet_pt_NOSYS 0.707
muonAllSegments_counts/nEOL 0.692
MS1Vtx_barrel_hits_ntot 0.632
jet_n_NOSYS 0.562
msvtx_jetMinDR_NOSYS 0.555
msvtx_nRPCMiddleStation 0.520
msvtx_nTrk 0.442
htmiss_y_NOSYS 0.436
MSVtx_MET_dphi_raw 0.429
msvtx_nMDTOuterStation 0.414
htmiss_x_NOSYS 0.409
MSeg_barrel_avg_dR 0.366
MS1Vtx_mindR_jetcut_raw 0.365
muonAllSegments_counts/nBOL 0.333
msvtx_nMDT 0.331
averageInteractionsPerCrossing 0.326
msvtx_trackHardMinDR_NOSYS 0.320
nBOL_NOSYS 0.319
msegUnAssoc_nSeg 0.314
msvtx_nMDTMiddleStation 0.313
MS1Vtx_maxclusE_raw 0.312
msvtx_caloClusterSumE 0.310
msvtx_caloClusterSumPt 0.308
msvtx_caloClusterSumPtScalar 0.307
MS1Vtx_avg_clusE_raw 0.304
MS1Vtx_rms_clusE_raw 0.299
MSeg_barrel_rms_dR 0.297
msvtx_el3ecal 0.293
muonAllSegments_counts/nBML 0.292
msvtx_phi_NOSYS 0.285
0.0
0.2
0.4
0.6
0.8
1.0
KS D, combined signal (equal weight per mass) vs background · tick = noise floor

Top 40 table · region BARREL

#	feature	D_equal	noise	D_pooled	D_min	D_max	n_bkg
1	met_significance_NOSYS	0.898	0.051	0.919	0.831	0.962	1,089
2	jet_pt_max_NOSYS	0.873	0.051	0.892	0.789	0.916	1,089
3	msegUnAssoc_counts/nEML	0.858	0.051	0.843	0.821	0.894	1,089
4	muonAllSegments_counts/nEML	0.848	0.051	0.828	0.799	0.901	1,089
5	msvtx_nRPCOuterStation	0.815	0.051	0.843	0.740	0.881	1,089
6	htmissScalar_NOSYS	0.798	0.051	0.818	0.798	0.872	1,089
7	nMSeg_ratio_EIEM	0.755	0.051	0.746	0.727	0.805	1,089
8	met_sumet_NOSYS	0.737	0.051	0.783	0.588	0.852	1,089
9	msvtx_nRPC	0.725	0.051	0.787	0.647	0.850	1,089
10	nEOL_NOSYS	0.710	0.051	0.651	0.580	0.844	1,089
11	msvtx_closestJet_pt_NOSYS	0.707	0.051	0.728	0.639	0.740	1,089
12	muonAllSegments_counts/nEOL	0.692	0.051	0.632	0.563	0.821	1,089
13	MS1Vtx_barrel_hits_ntot	0.632	0.051	0.727	0.405	0.834	1,089
14	jet_n_NOSYS	0.562	0.051	0.611	0.458	0.691	1,089
15	msvtx_jetMinDR_NOSYS	0.555	0.051	0.583	0.510	0.622	1,089
16	msvtx_nRPCMiddleStation	0.520	0.051	0.611	0.314	0.691	1,089
17	msvtx_nTrk	0.442	0.051	0.458	0.418	0.475	1,089
18	htmiss_y_NOSYS	0.436	0.051	0.437	0.374	0.470	1,089
19	MSVtx_MET_dphi_raw	0.429	0.051	0.407	0.332	0.620	1,089
20	msvtx_nMDTOuterStation	0.414	0.051	0.565	0.080	0.710	1,089
21	htmiss_x_NOSYS	0.409	0.051	0.421	0.363	0.458	1,089
22	MSeg_barrel_avg_dR	0.366	0.051	0.497	0.101	0.645	1,089
23	MS1Vtx_mindR_jetcut_raw	0.365	0.051	0.391	0.307	0.432	1,089
24	muonAllSegments_counts/nBOL	0.333	0.051	0.425	0.151	0.517	1,089
25	msvtx_nMDT	0.331	0.051	0.500	0.161	0.687	1,089
26	averageInteractionsPerCrossing	0.326	0.051	0.324	0.307	0.367	1,089
27	msvtx_trackHardMinDR_NOSYS	0.320	0.051	0.346	0.273	0.393	1,089
28	nBOL_NOSYS	0.319	0.051	0.408	0.142	0.508	1,089
29	msegUnAssoc_nSeg	0.314	0.051	0.175	0.113	0.652	1,089
30	msvtx_nMDTMiddleStation	0.313	0.051	0.444	0.196	0.573	1,089
31	MS1Vtx_maxclusE_raw	0.312	0.051	0.314	0.275	0.332	1,089
32	msvtx_caloClusterSumE	0.310	0.051	0.312	0.282	0.343	1,089
33	msvtx_caloClusterSumPt	0.308	0.051	0.308	0.284	0.346	1,089
34	msvtx_caloClusterSumPtScalar	0.307	0.051	0.308	0.284	0.347	1,089
35	MS1Vtx_avg_clusE_raw	0.304	0.051	0.310	0.252	0.333	1,089
36	MS1Vtx_rms_clusE_raw	0.299	0.051	0.309	0.261	0.331	1,089
37	MSeg_barrel_rms_dR	0.297	0.051	0.414	0.133	0.584	1,089
38	msvtx_el3ecal	0.293	0.051	0.300	0.251	0.331	1,089
39	muonAllSegments_counts/nBML	0.292	0.051	0.371	0.097	0.477	1,089
40	msvtx_phi_NOSYS	0.285	0.051	0.285	0.279	0.292	1,089

* = D_equal below its noise floor (not distinguishable from no separation)


Top 40 · region ALL
met_significance_NOSYS 0.882
jet_pt_max_NOSYS 0.856
htmissScalar_NOSYS 0.789
met_sumet_NOSYS 0.702
msvtx_closestJet_pt_NOSYS 0.689
msvtx_nMDT 0.560
msvtx_nTrk 0.545
jet_n_NOSYS 0.525
msvtx_jetMinDR_NOSYS 0.504
msvtx_nMDTMiddleStation 0.483
msvtx_nTGCOuterStation 0.468
msvtx_nMDTOuterStation 0.465
MS1Vtx_endcap_hits_ntot 0.462
msvtx_nTGC 0.451
msvtx_nTGCExtendedStation 0.448
htmiss_y_NOSYS 0.418
MS1Vtx_avg_clusE_raw 0.414
msvtx_caloClusterSumE 0.414
MS1Vtx_rms_clusE_raw 0.413
MS1Vtx_maxclusE_raw 0.411
msvtx_chiSquared 0.405
nMSeg_ratio_BIBM 0.404
msvtx_caloClusterSumPtScalar 0.391
msvtx_caloClusterSumPt 0.391
MSVtx_MET_dphi_raw 0.389
msvtx_nMDTExtendedStation 0.388
htmiss_x_NOSYS 0.386
msvtx_el3ecal 0.384
msvtx_nTGCMiddleStation 0.381
msvtx_nRPCOuterStation 0.377
msvtx_nRPC 0.374
averageInteractionsPerCrossing 0.370
msvtx_nCaloClusters_Egt300MeV_NOSYS 0.367
msvtx_nCaloClusters_NOSYS 0.365
msvtx_nRPCMiddleStation 0.362
msvtx_el2ecal 0.359
MS1Vtx_maxclustime_raw 0.356
msvtx_el4ecal 0.349
msvtx_el1hcal 0.345
MS1Vtx_rms_clustime_raw 0.337
0.0
0.2
0.4
0.6
0.8
1.0
KS D, combined signal (equal weight per mass) vs background · tick = noise floor

Top 40 table · region ALL

#	feature	D_equal	noise	D_pooled	D_min	D_max	n_bkg
1	met_significance_NOSYS	0.882	0.018	0.902	0.814	0.938	36,059
2	jet_pt_max_NOSYS	0.856	0.018	0.876	0.782	0.913	36,059
3	htmissScalar_NOSYS	0.789	0.018	0.792	0.789	0.868	36,059
4	met_sumet_NOSYS	0.702	0.018	0.753	0.563	0.856	36,059
5	msvtx_closestJet_pt_NOSYS	0.689	0.018	0.703	0.640	0.719	36,059
6	msvtx_nMDT	0.560	0.018	0.666	0.366	0.783	36,059
7	msvtx_nTrk	0.545	0.018	0.548	0.502	0.569	36,059
8	jet_n_NOSYS	0.525	0.018	0.572	0.422	0.659	36,059
9	msvtx_jetMinDR_NOSYS	0.504	0.018	0.525	0.480	0.548	36,059
10	msvtx_nMDTMiddleStation	0.483	0.018	0.573	0.321	0.670	36,059
11	msvtx_nTGCOuterStation	0.468	0.018	0.505	0.371	0.532	36,059
12	msvtx_nMDTOuterStation	0.465	0.018	0.543	0.326	0.634	36,059
13	MS1Vtx_endcap_hits_ntot	0.462	0.018	0.511	0.335	0.547	36,059
14	msvtx_nTGC	0.451	0.018	0.498	0.354	0.527	36,059
15	msvtx_nTGCExtendedStation	0.448	0.018	0.490	0.351	0.519	36,059
16	htmiss_y_NOSYS	0.418	0.018	0.422	0.364	0.447	36,059
17	MS1Vtx_avg_clusE_raw	0.414	0.018	0.421	0.396	0.429	36,059
18	msvtx_caloClusterSumE	0.414	0.018	0.415	0.411	0.420	36,059
19	MS1Vtx_rms_clusE_raw	0.413	0.018	0.420	0.399	0.428	36,059
20	MS1Vtx_maxclusE_raw	0.411	0.018	0.414	0.394	0.419	36,059
21	msvtx_chiSquared	0.405	0.018	0.490	0.494	1.000	36,059
22	nMSeg_ratio_BIBM	0.404	0.018	0.410	0.381	0.427	36,059
23	msvtx_caloClusterSumPtScalar	0.391	0.018	0.392	0.378	0.403	36,059
24	msvtx_caloClusterSumPt	0.391	0.018	0.392	0.377	0.403	36,059
25	MSVtx_MET_dphi_raw	0.389	0.018	0.368	0.292	0.536	36,059
26	msvtx_nMDTExtendedStation	0.388	0.018	0.358	0.344	0.468	36,059
27	htmiss_x_NOSYS	0.386	0.018	0.401	0.356	0.444	36,059
28	msvtx_el3ecal	0.384	0.018	0.389	0.368	0.403	36,059
29	msvtx_nTGCMiddleStation	0.381	0.018	0.435	0.318	0.471	36,059
30	msvtx_nRPCOuterStation	0.377	0.018	0.419	0.306	0.478	36,059
31	msvtx_nRPC	0.374	0.018	0.417	0.303	0.476	36,059
32	averageInteractionsPerCrossing	0.370	0.018	0.374	0.338	0.396	36,059
33	msvtx_nCaloClusters_Egt300MeV_NOSYS	0.367	0.018	0.368	0.357	0.383	36,059
34	msvtx_nCaloClusters_NOSYS	0.365	0.018	0.366	0.352	0.383	36,059
35	msvtx_nRPCMiddleStation	0.362	0.018	0.401	0.293	0.454	36,059
36	msvtx_el2ecal	0.359	0.018	0.366	0.334	0.375	36,059
37	MS1Vtx_maxclustime_raw	0.356	0.018	0.360	0.330	0.372	36,059
38	msvtx_el4ecal	0.349	0.018	0.355	0.309	0.379	36,059
39	msvtx_el1hcal	0.345	0.018	0.348	0.330	0.357	36,059
40	MS1Vtx_rms_clustime_raw	0.337	0.018	0.344	0.292	0.355	36,059

* = D_equal below its noise floor (not distinguishable from no separation)